ANALOG INTEGRATED CIRCUITS

Using the Differential Amp designed in Lab 3, this lab will focus on developing higher gain two-Stage amplifier. You will simulate the amplifier with frequency compensation to increase the system stability. You may change the parameters of the amplifier designed in previous lab to achieve the requested performance parameters here.

Design Specifications:

- Power Supply: 2V; Power Consumption: 1mW maximum
- CL: 2 pF • DC Gain: 70 dB minimum; output Swing> 1.3 Vpp
- Design for maximum BW and unity gain frequency
- RC compensation for stability: Phase Margin: 60 degrees minimum
- Use biasing circuits discussed in the class to properly bias your devices.

Hand Calculations:

$K_n' = 2.7475 \times 10^{-4}\ A/V^2$

$K_p' = 0.9575 \times 10^{-4}\ A/V^2$

$l = 0.3\ \mu m$

But $2l = 0.6 = 600\ nm$

$C_L = 2\ pf$

O/P Swing > 1.3 V_{pp}

DC Gain > 70 dB

Let $I_D = 10\ \mu A$

∴ first transistor from left

$I_d = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{ov})^2$

Assume $V_{ov} = 0.2\ V$

$\therefore 10\ \mu A = \frac{1}{2} \times 0.9575 \times 10^{-4}$

$\therefore W_1 = 6.3\ \mu m$

For transistor 2

$W_2 = 6.3\ \mu m \longrightarrow$ Current mirror

for transistor 3 & 4 Nmos

$I_D = \frac{1}{2} K_n \frac{W}{L} (V_{GV})^2$

$W = 2.175\ \mu m$

for transistor 5

$W = 6.33\ \mu m$

for transistor 6

Part of Cascode

$L = 4.8\ \mu m$

$$W = \frac{2 I_D \times L}{K_n \times V_{ov}^2} = \frac{2 \times 10\mu \times 1.2\mu}{0.9475 \times 10^{-4} (0.2)^2}$$

$W = 6.33\ \mu m$

for transistor 7 & 8 (PMOS) $W = 6.32\mu$

for PMOS transitor 9 : $L = 2.4\ \mu$

$W = 0.44\ \mu m$

for stage 1 dif amp

1st NMOS towards gnd

$\therefore I_{ov} = \frac{1}{2} \frac{W}{L} K_n (V_{ov})^2$

$W = 15.8\ \mu m$

for other 2 NMOS

$W = \frac{15.8}{2} = 7.9\ \mu m$

for 2 PMOS transistors:

$W = 48.48\ \mu m$

for stage 2:

$I = 880\ \mu A$

$\therefore I_D = \frac{1}{2} \frac{W}{L} K_p (V_{OV})^2$

$880\ \mu A = \frac{1}{2} \frac{W}{L} K_p (0.2)^2$

$W = 41.99\ \mu m$

For gain

$$\text{Gain} = \frac{I}{2V_{ov}}\left(\frac{I_{cascode}}{V_{ov}(\lambda^2 p)} \,\|\, \frac{I_{cascode}}{V_{ov} I (\lambda_n)^2}\right)$$

$$= \frac{I}{V_{ov}(\lambda_n + \lambda_p)}\left\{\frac{I}{V_{ov}}\left(\frac{1}{\lambda^2 p} \,\|\, \frac{1}{\lambda^2 n}\right)\right.$$

$\lambda_n + \lambda_p = 0.25\,u$

$\lambda^2 p = 0.161 = \lambda_n^2$

$$\text{Gain} = \frac{2}{0.2(0.25}\left\{\frac{1}{0.2}\left(\frac{1}{0.161\,2} \,\|\, \frac{1}{0.16\,2}\right)\right.$$

$A_v(dB) = 81.7\,dB$

For diff amp (stage 1)

$$r_{op_1} = \frac{1/\lambda p}{I_0} = 250\,K\Omega$$

$$r_{on_1} = \frac{1/\lambda_n}{I_0} = 200\,K\Omega$$

Stage 2

$$r_{op2} = \frac{1/\lambda p}{I_d} = 62.5\ K\Omega$$

$$r_{on2} = 50\ K\Omega$$

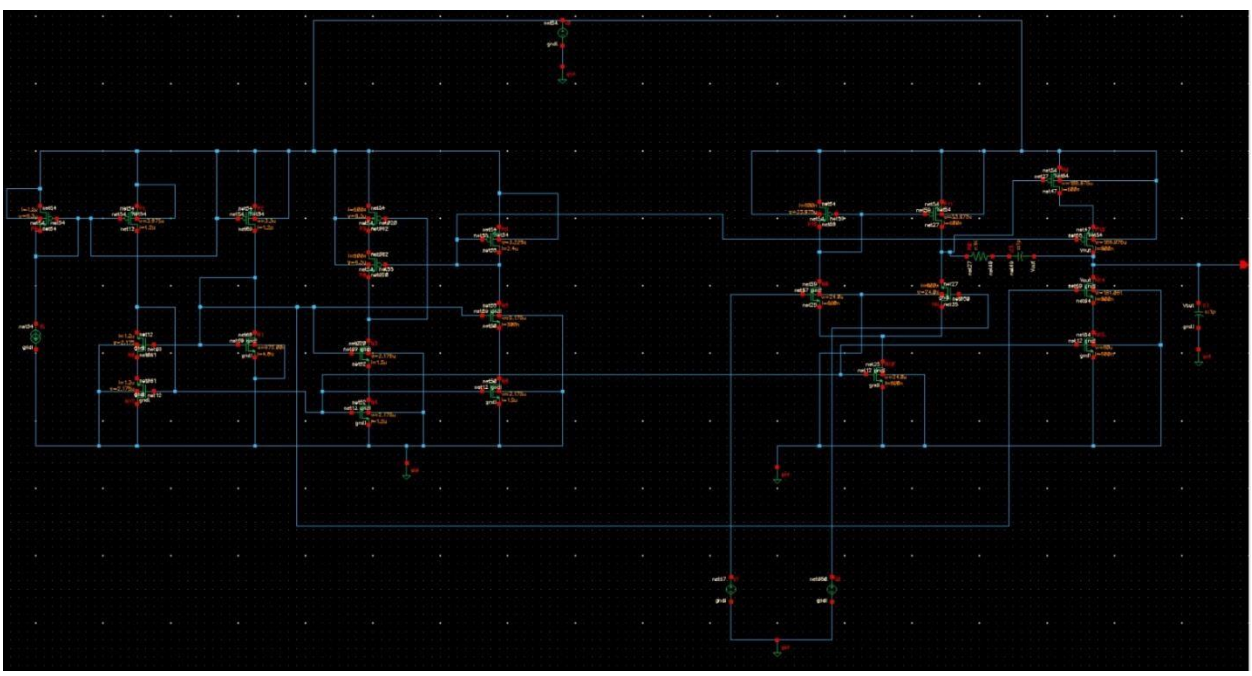
Setting time:

Time constant RC

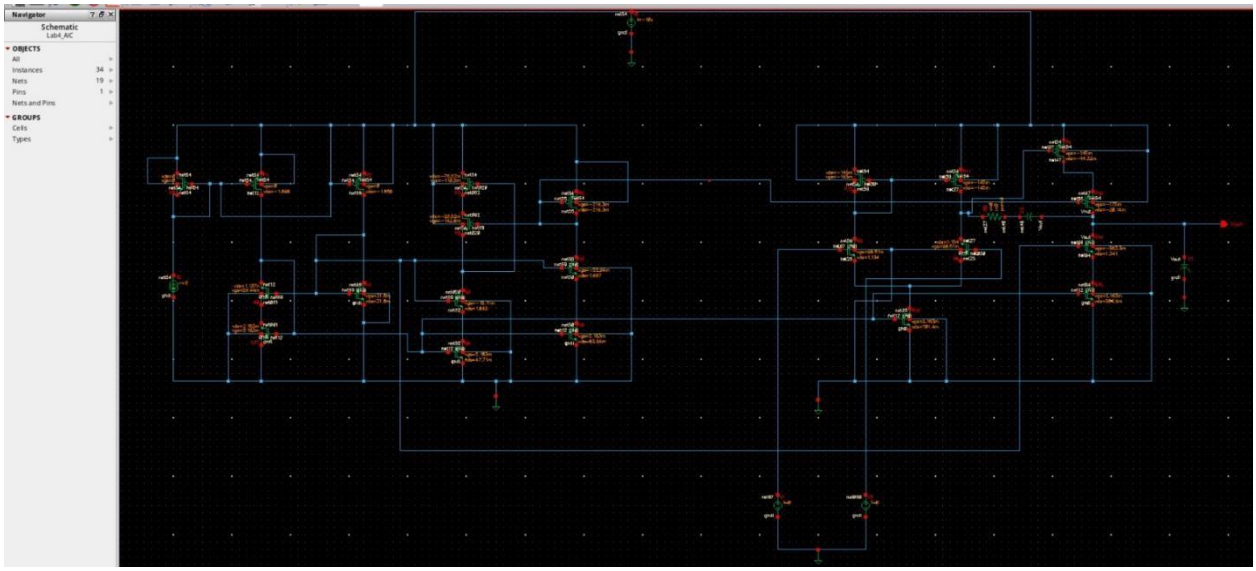
$$R = \frac{1}{\lambda I_p}$$

$$\therefore\ T_s = 0.647\ ns$$

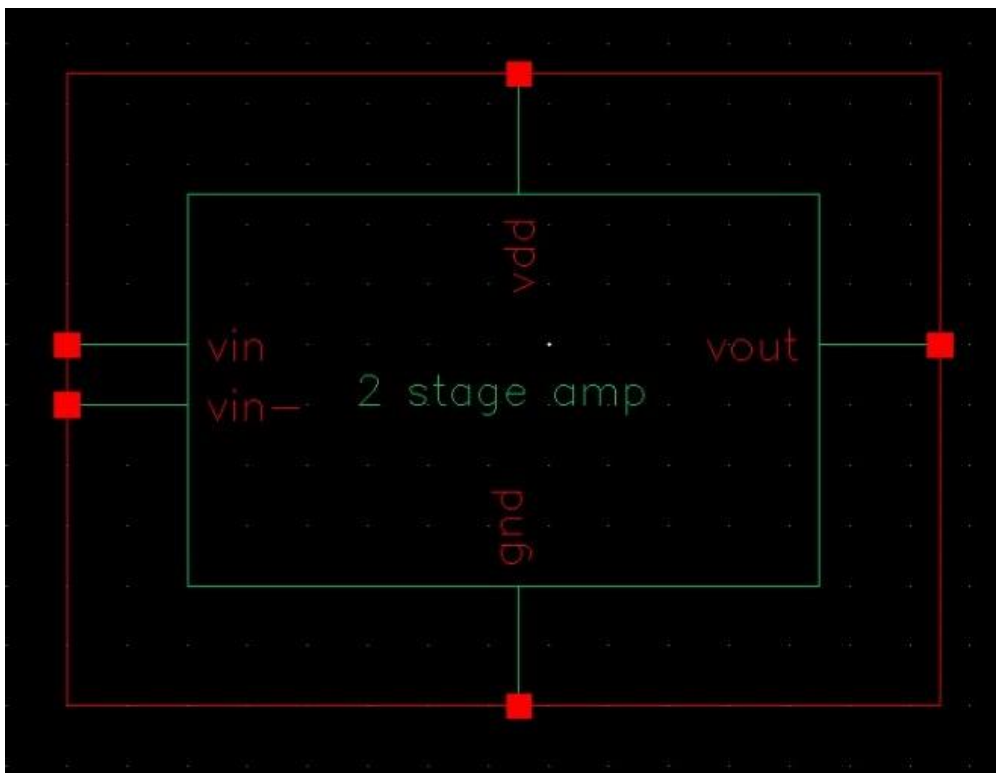
Schematic:



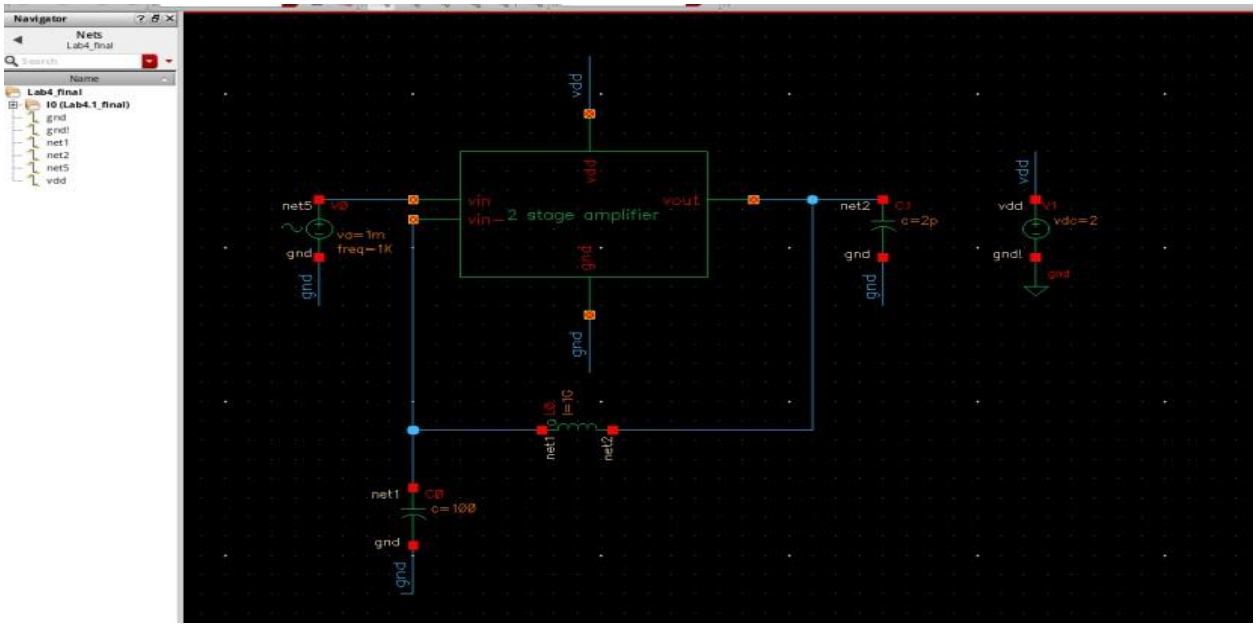
DC operating points:



Symbol:

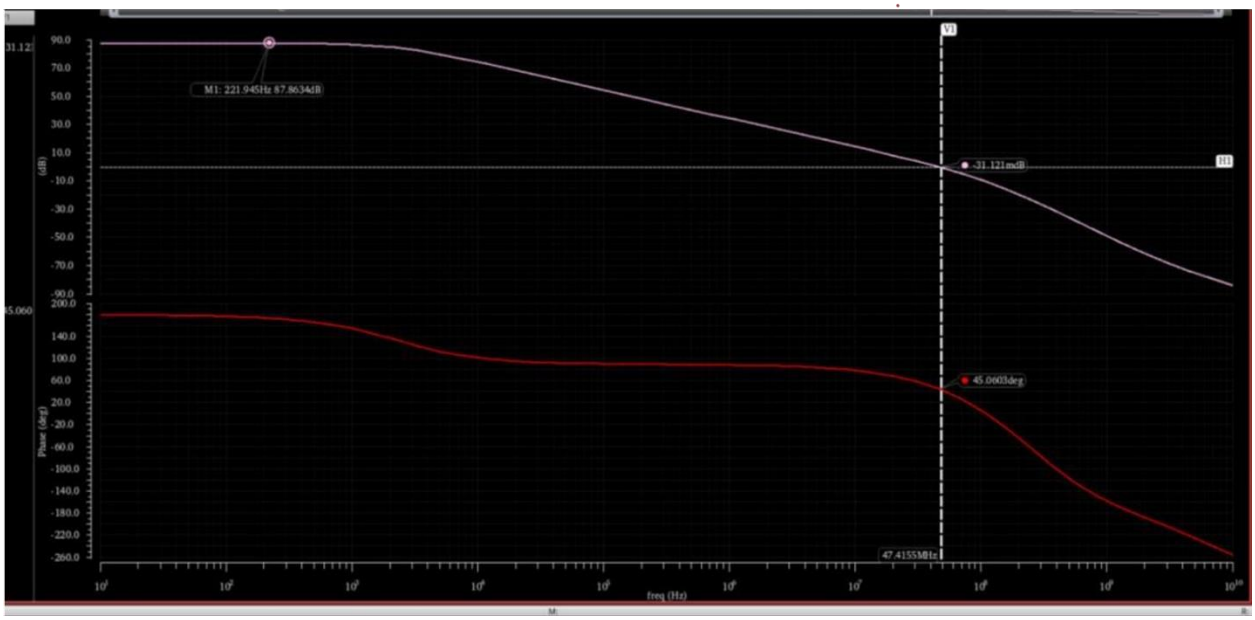


2 stage amplifier circuit:



AC Simulations:

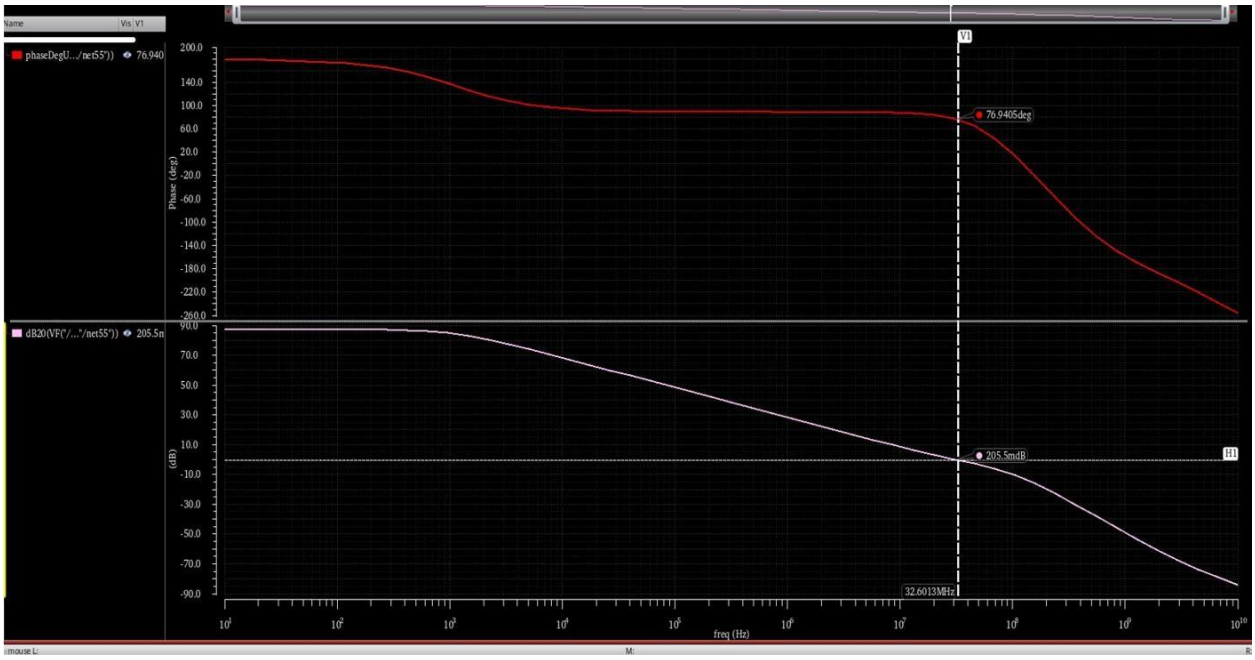
Gain and Phase vs Freq



Gain should be greater than 70dB

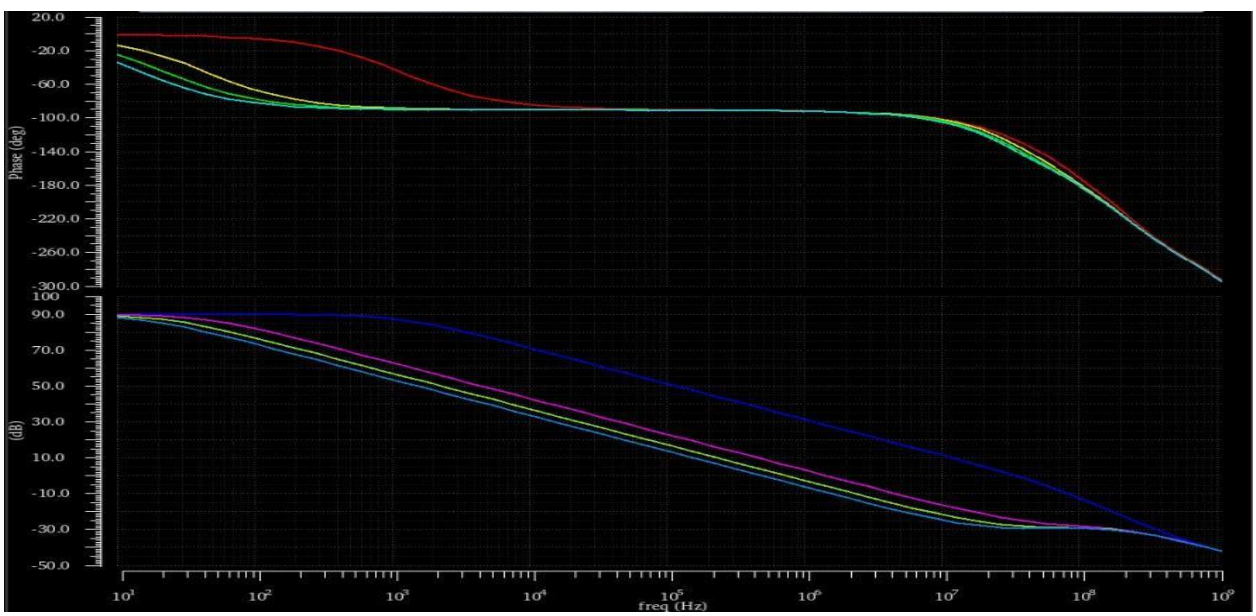
Gain from simulations: **87.8 dB**

Phase Margin



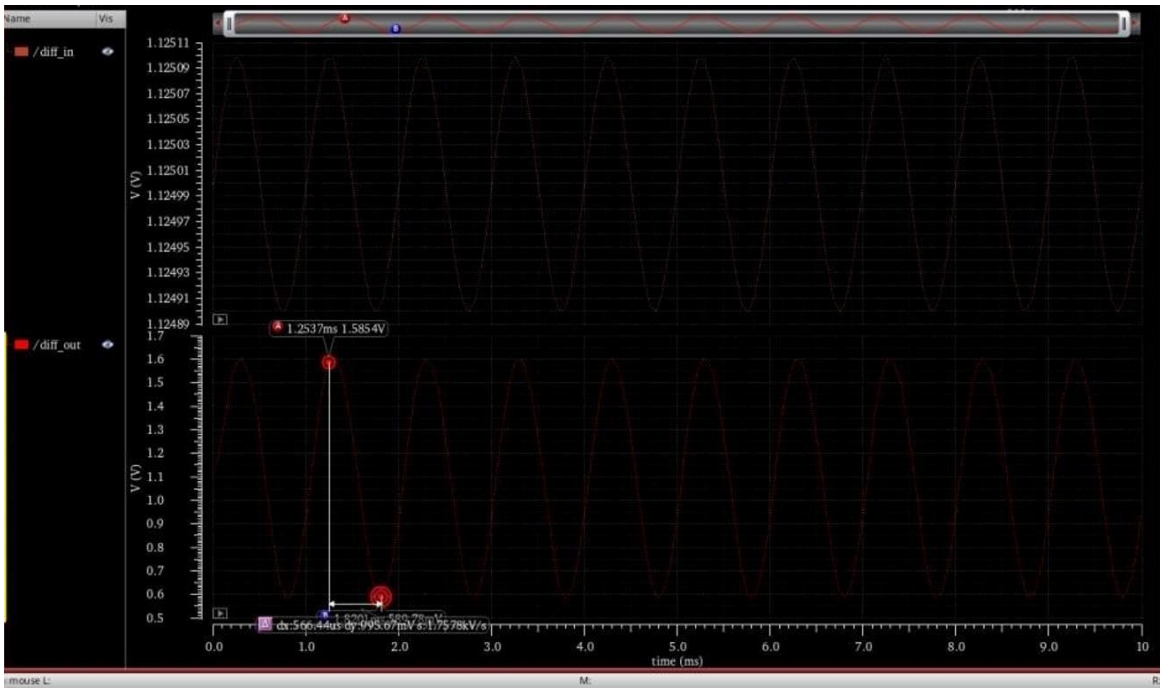
Phase Margin is around **76.9 deg Parametric**

analysis:



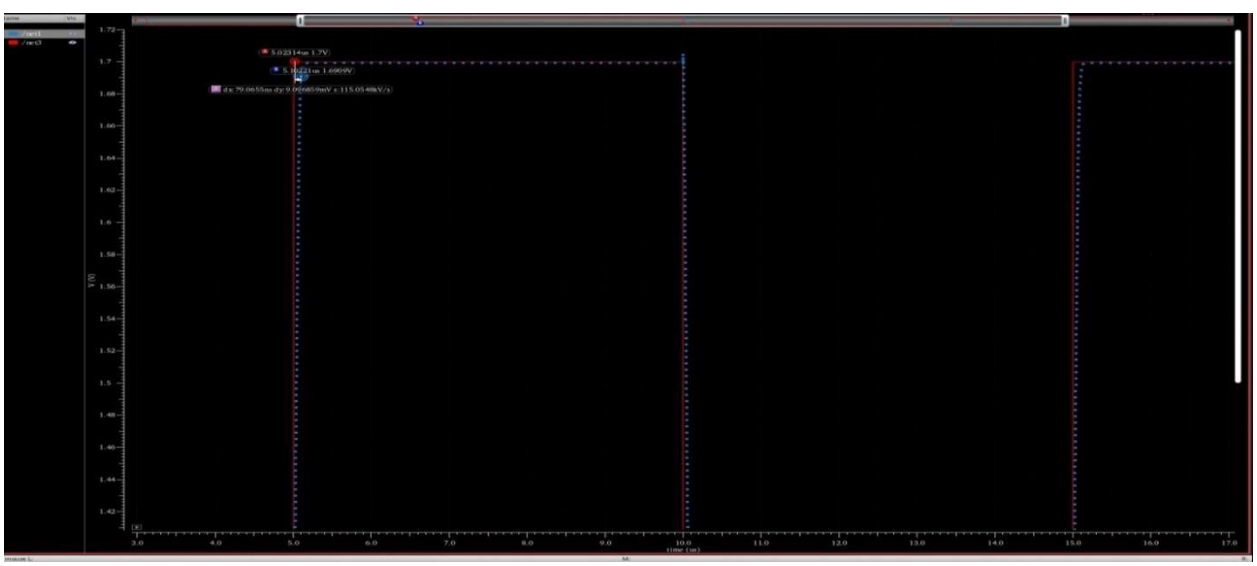
The sweep value is -0.764dB for 12.76 MHz

Output Swing graph:

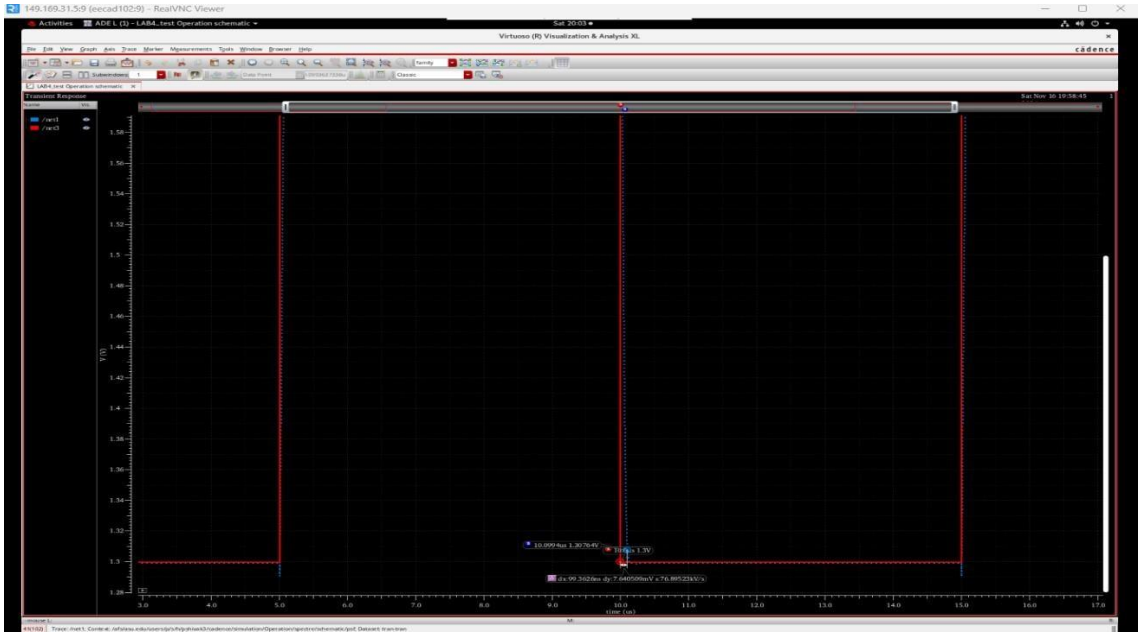


Output swing is Vmax-Vmin = 1.6052V – 0.2901 V = **1.3152 Vpp Transient**

Simulations:



Transient Analysis of Raise time



Transient Analysis of Fall Time **Task 3:**

- What are the sources of discrepancy?
  a) To achieve the necessary phase margin, the compensation capacit or was slightly increased.
  b) To ensure saturation and appropriate biasing, cascode transistor l engths were modified. c) Although the nondominant pole was not as calculated, the phase margin was enhanc ed by coupling capacitance, and the zero was positioned appropriate ly for signal attenuation.
- Difficulties in fulfilling performance standards:
  Although it was challenging, it was nearly possible to achieve an output swing above 1.3 V without distortion.
  A maximum output swing of 1.3152Vpp was attained.

1. What would you change to achieve higher bandwidth and what are the trade offs
   While decreasing the compensation capacitor (Cc) preserves bandwidth at the expense of phase margin, increasing it increases phase margin but decreases bandwidth.

Conclusion: 1. Phase margin: 76.9 degrees, amplifier gain:

87.8 dB.

2. The findings of manual calculations and simulations closely match, and they are accurate.